What a smartphone protocol can establish about urban noise:
three negative results and an agent-based exposure model for Hanoi

Laurian Jamin Lucas Zborowski

ISIMA, Clermont-Ferrand, France, COSMOS Lab, VinUniversity, Hanoi

laurian@laurian-jamin.fr

Abstract

Hanoi is dense, motorcycle-dominated and has no routine noise monitoring, and professional instrumentation is out of reach of most local budgets. We report what a noise-mapping protocol built from three consumer smartphones and open data does and does not establish, over 363 field measurements taken across three contrasted urban typologies between 10 June and 22 July 2026. Evaluated under three cross-validation protocols of increasing severity, with a 2 000-replicate block bootstrap, our results are three negative ones. **(i)** A three-parameter line-source attenuation kernel outperforms every learned model we built, including a physics-plus-residual hybrid we had ourselves recommended: the ranking of seven models *inverts* between a permissive spatial split and two strict ones. **(ii)** Cross-city transfer from a 61 000-point Ugandan corpus fails, not by an offset, but by carrying no usable information at all, while one locally fitted distance term reaches $R^2 = 0.240$. **(iii)** Vehicle counts extracted from 147 matched traffic videos yield exactly zero emission coefficients for motorcycles and cars, as density and as line-crossing flow alike. We deliver the surviving model as a 40 m exposure map driving an agent-based simulation in which *receiver* agents, not emitter agents, accumulate a noise dose, a design forced by the absence of any per-street data that emitter agents could be calibrated or verified against.

1 Introduction

Vietnam has no national strategic noise mapping programme and no statutory noise action plan (Nguyen et al., 2025a), and class 1 or class 2 sound level meters and commercial propagation software sit outside most local research budgets. This is exactly the setting in which low-cost participatory noise mapping is proposed. It is also a setting in which the claims such a method can support are rarely tested.

We ran a three-month campaign in Hanoi and then evaluated it adversarially. Our contribution is not a noise map of Hanoi, our instrument cannot support one, but a set of **boundary conditions on low-cost urban noise modelling**, each obtained from an experiment designed to succeed.

The agent-based component is where the constraint bites hardest and is, we believe, of independent interest: with no per-street flows, no modal shares, no speeds and no signal cycles, an emitter-agent traffic simulation would be neither calibratable nor verifiable. We therefore invert the usual construction and put the agents on the *receiving* side of a measured field.

2 Data and protocol

363 measurements were collected across three deliberately contrasted typologies , a new peri-urban development (Ocean Park, $n = 184$), a historic dense quarter (Hoan Kiem lake, $n = 99$) and a transport corridor (Vinh Tuy, $n = 80$), using ODK Collect against a KoboToolbox server, with a 10 m GPS accuracy gate and a recorded outdoor distance to the road. 147 timestamped traffic videos were recorded alongside (median video-to-measurement offset 15 s).

The quantity, stated precisely. Each record is one A-weighted, SLOW-weighted level over a 20–30 s stationary observation, which we denote $L_{A,25\text{s}}$ and never $L_{A\text{eq}}$, L_{den} or L_{night}. The three handsets were cross-calibrated *against one another* and against no standard, so a bias common to all three is invisible in the data by construction. Consumer smartphone measurement is known to depart from reference instruments by several decibels in a handset-, OS- and level-dependent way that is not a constant offset (Kardous and Shaw, 2014; Murphy and King, 2016). Accordingly:

- contrasts between places, hours and typologies are supported, because a common bias cancels in a difference;
- absolute levels are reported with a bounded bias interval (+3.0 to +7.3 dB) estimated against instrumented Vietnamese campaigns (Phan et al., 2010; Gelb and Apparicio, 2019) and never applied;
- **no compliance claim is made.** QCVN 26:2010 (Ministry of Natural Resources and Environment of Viet Nam, 2010) thresholds appear only as descriptive colour bands. A WHO $L_{\text{den}}/L_{\text{night}}$ comparison present in an earlier version of this work was withdrawn in full.

3 Evaluation framework

Eight models are scored on *identical folds* under three protocols of increasing severity: 600 m spatial block cross-validation (17 blocks); buffered leave-one-out with

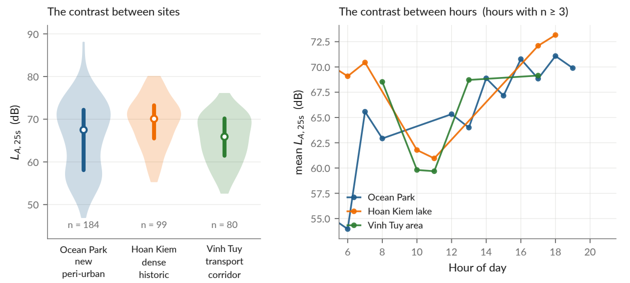

Figure 1: The 363 measurements: spatial layout, level distribution and temporal coverage across the three typologies.

Table 1: R^2 under the three protocols, 95 % block-bootstrap intervals, $n = 363$. Reference protocol in the middle column. Best per column in bold.

Model	**Block-CV** 600 m	**Buff. LOO** 300 m	**LOSO**
Mean by (site, hour)	−0.008	−0.419	−0.058
$\log d_{\mathrm{road}}$ (2 par.)	0.221	0.200	0.189
Physical (3 par.)	0.255	**0.246**	**0.222**
LightGBM v1 (6 feat.)	0.304	0.137	0.029
LightGBM v2 (8 feat.)	0.332	0.099	−0.035
Hybrid (physics + ML)	**0.395**	0.123	0.035
Hybrid, restricted	0.378	0.144	0.106

a 300 m exclusion radius; and leave-one-site-out. Confidence intervals come from a 2 000-replicate block bootstrap.

The middle protocol is our **reference**, because its exclusion radius equals the radius over which the morphological features are aggregated. This is not a detail. An earlier version of this work reported $R^2 = 0.45$ under a grouping on ∼110 m cells, smaller than the 300 m feature radius; that protocol leaked between folds (Roberts et al., 2017) and the figure was withdrawn. The delivered model is now selected *by code*, best score under the reference protocol among candidates fixed in advance, so that a published map cannot silently inherit a model that only wins on a permissive split.

4 Result 1: geometry beats learning

The delivered model treats each road class as an incoherent line source, whose intensity falls as $1/d$ rather than $1/d^2$:

$$E(x) = \frac{A_{\mathrm{hw}}}{\max(d_{\mathrm{hw}}, d_0)} + \frac{A_{\mathrm{res}}}{\max(d_{\mathrm{res}}, d_0)} + B,$$

$L(x) = 10 \log_{10} E(x)$, with d_{hw} and d_{res} the distances to the nearest major axis and to the nearest local street, separated by OSM `highway` tag, three non-negative coefficients and $d_0 = 5$ m.

Read Table 1 by column and the ranking reverses (Fig. 2). Every elaboration we added improves the permissive protocol and degrades the strict ones, monotonically, the signature of capacity fitting spatial autocorrelation rather than physics. Three readings follow.

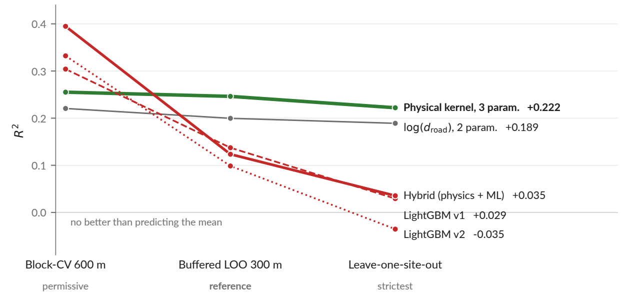

Figure 2: The inversion. Same models, same folds; the ordering reverses almost exactly as the split begins to test generalisation.

The morphological aggregates have negative marginal value. Adding built-area ratio, road density and intersection count to distance-to-road degrades prediction by 0.07 R^2 under block-CV, 0.16 under buffered LOO and 0.18 under leave-one-site-out. A 300 m disk is large enough to average away the canyon geometry that governs propagation and small enough to be strongly autocorrelated between neighbouring points: it supplies variance without supplying information.

What survives in the learned model is time, not space. The gap between the full model and the morphology-only ablation under block-CV (0.304 vs 0.153) is carried by hour-of-day. Under leave-one-site-out the time-only ablation is itself negative (−0.139): it does not help predict an unmeasured location, which is what a map is for.

The hybrid was built and rejected, not deferred. We proposed the physics-plus-learned-residual architecture ourselves as the constructive answer to our own negative results. Built and evaluated on identical folds, the residual gains $\Delta R^2 = +0.140$ under block-CV and loses −0.123 and −0.187 under the two strict protocols. Constraining its capacity recovers part of the loss, and the direction of that trade is the diagnosis: what the unconstrained residual learns is the spatial arrangement of our points, not physics the core is missing. The residual is computed at every run and deliberately not applied.

A ceiling, so that $R^2 = 0.25$ can be read. Two measurements within 40 m of each other *at the same hour* must be predicted identically by any spatial model, so their disagreement is variance no spatial model can explain. Over 87 such pairs the mean absolute difference is 5.44 dB, implying $\sigma \approx 4.8$ dB against a total variance of 51 dB^2, a ceiling near $R^2 \approx 0.54$. The delivered model reaches about half of what is attainable, not a tenth of it. (Pairs at the same hour may fall on different days, so this σ carries day-to-day variation: it is an order of magnitude, not a bound.)

5 Result 2: transfer fails, and not by an offset

Following the methodology we reproduce (Nsumba et al., 2026), we pretrained a morphology-to-level model on the Uganda 61K corpus and applied it to Hanoi. The feature set was deliberately reformulated to be invariant

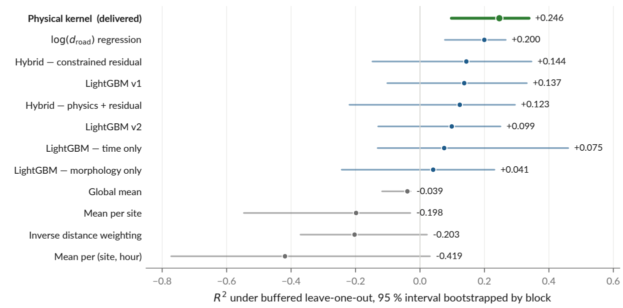

Figure 3: Every model under the reference protocol with its 95 % interval. The physical kernel is the only model whose interval excludes zero under all three protocols, and it is not separated from the two-parameter distance regression.

Table 2: Uganda-pretrained models evaluated on the 363 Hanoi points. Re-run 24 August 2026 from the versioned boosters.

Pretrained	as delivered	mean diff. removed	rescaled ($= r^2$)	r
Uganda 61K (v1)	−15.86	−2.14	+0.141	−0.376
Uganda inv. (v2)	−8.17	−0.95	+0.006	+0.076
Fitted locally on the same points:				
$\log d_{\text{road}}$	$R^2 = +0.240$ (in sample)			
Physical, buff. LOO	$R^2 = +0.246$			

to OSM mapping conventions after a first version proved sensitive to them (Kampala is mapped as individual buildings, Barcelona as whole blocks).

Three readings of Table 2, and the third is the one that matters. **It is not a calibration offset**: the Kampala model predicts about 26 dB below Hanoi, and removing that difference leaves R^2 negative, so a recalibrated transfer would not work either. **v1 transfers anti-information**: its correlation with Hanoi is negative, it ranks quiet and loud the wrong way round. **v2 removes the convention artefact and leaves nothing**: $r = +0.076$, and even granted a free bias and a free slope its ceiling is $R^2 = 0.006$. Not wrong, empty.

Against which, one locally fitted distance term on the same 363 points reaches $R^2 = 0.240$. **Local measurement is not a refinement of transfer; it is a prerequisite.** We attribute the failure to fleet composition and horn behaviour (Nguyen et al., 2025b), to the same morphological descriptor denoting different urban forms in different cities, and to target quantities that are not exchangeable across studies.

6 Result 3: video counts carry no acoustic signal

Regressing measured acoustic *energy* on vehicle counts, $E = E_{\text{bg}(s)} + \sum_c n_c e_c$ with $e_c \geq 0$, returns exactly zero for motorcycles and cars. When we traced this to the choice of quantity, a per-frame count is a *stock*, while emission is driven by *flow*, and the two decouple exactly in congestion, we implemented our own recommendation, re-processing all 147 videos with multi-object tracking and line-crossing counts.

Flow does not rescue it. Pooled, total flow against

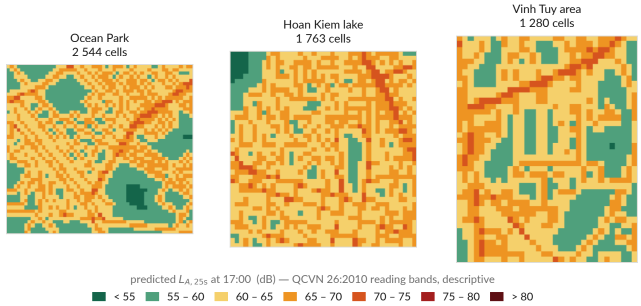

Figure 4: The delivered 40 m grid, 5 587 cells over three zones × 17 hours, restricted to the sampled envelope (three sites plus 400 m). Colour bands are QCVN 26:2010 references, shown descriptively.

level gives $r = -0.109$ and motorcycle flow $r = -0.004$. The two obstacles we did not remove are the explanation: speed is unobservable without a ground homography and dominates emission above 30–40 km/h (Kephalopoulos et al., 2014), and distance is discarded because the field of view is not georeferenced.

> Instrument for (Q, v) *and* georeference the field of view. Obtaining Q alone buys a coherent traffic statistic and no acoustic calibration.

We state a caveat on our own negative result. Our detector was never validated against manual counts, and it returns a modal split of 57 % cars against 36 % motorcycles, in Hanoi, where reality is roughly the inverse. This result should therefore be read as *"the chain could not be made to work"* rather than as a finding about acoustics.

7 Receiver agents over a measured field

The surviving model is exported as a 40 m grid (Fig. 4) and drives an agent-based simulation in GAMA. The design decision is the paper's fourth contribution, and it is a negative one applied constructively.

Emitter agents were rejected. Motorcycle and car agents need real per-street flows, fleet composition, speeds and signal cycles. We have none of these for Hanoi, Result 3 is precisely the failure of our attempt to recover them, so an emitter simulation would be neither calibratable against data nor verifiable against anything.

Receiver agents were adopted. Agents move through the area and accumulate a *noise dose*: the level of the cell crossed, times the time spent in it. The input is a map that was measured and validated; nothing about the sources is invented. What emerges at population level, the distribution of daily exposure, is what a map alone cannot give.

Against 360 matched measurement points the simulated field has mean error −1.24 dB, MAE 5.30 dB and RMSE 6.49 dB. It reproduces the spatial and temporal ordering and does not reproduce the spread, which is expected: features aggregated over 300 m cannot resolve a facade/courtyard contrast. One correction is worth recording, because it is easy to get wrong, the traffic multiplier must scale the *traffic share* of the energy and never the background; applied to total energy it double-counts the background and inflated the apparent benefit of pedestrianisation from 3.5 dB to 7.0 dB.

Mobile receiver agents with daily journeys ("tier 2") are specified and **not yet implemented**; they are the natural next increment.

8 Limitations

Two limitations are not fixable by code and bound everything above. **There is no absolute reference**: three handsets cross-calibrated against each other and against no standard. **Three sites are three typologies**, the effective number of independent morphological configurations is close to three whatever the number of points, and three is not a basis for claiming coverage of a city. In line with area-of-applicability reasoning (Meyer and Pebesma, 2021) we therefore restrict every published map to the sampled envelope, irrespective of score, and we withdrew an earlier map covering a district in which nothing was measured.

Three further limits are open: the headline ranking is not statistically separated (physical $0.246\,[0.097, 0.339]$ against $\log d_{\mathrm{road}}$ $0.200\,[0.077, 0.266]$, Fig. 3); the vehicle detector is unvalidated; and a -5 dB discontinuity in the level distribution on 27 June 2026, coinciding with a change of form version, is spatially structured and has no term in any of our models; it is not diagnosed here.

9 Conclusion and future work

At our sampling density and typological coverage, the predictable part of the spatial variation in urban noise is the geometric divergence of sound from the road network. A three-parameter line-source model captures it more robustly than any learned model we built, including a hybrid using that same model as its base; morphology aggregated over 300 m adds no measurable value under any protocol. We recommend that a distance-to-source physical baseline be reported as a mandatory reference in every morphology-based noise mapping study, under a spatial split whose block size exceeds the feature aggregation radius.

The indicated continuation is a real propagation kernel, CNOSSOS-EU via NoiseModelling (Kephalopoulos et al., 2014; Bocher et al., 2019), which models what a distance law cannot: diffraction around buildings and first- and second-order reflections. If a two-parameter distance term already beats a six-variable gradient-boosting model, a physical propagation core corrected by a locally learned residual is the architecture the data points at.

Data and code availability. The 363 measurements, the 147 video counts, the survey forms, the full pipeline, the GAMA model and the evaluation framework are released at `https://github.com/Colauz/noise-modelling-hanoi` (code MIT, data CC-BY-4.0). Traffic videos and raw survey exports are not distributed, for the reasons stated there.

Acknowledgements. This work was carried out at COSMOS Lab, VinUniversity, Hanoi, under the supervision of Doanh Nguyen-Ngoc. We thank Nguyen Thanh Quang (VinUniversity) for his contribution to the field campaign and for running the distribution of the field application.

References

Erwan Bocher, Gwenaël Guillaume, Judicaël Picaut, Gwendall Petit, and Nicolas Fortin. NoiseModelling: an open source GIS-based tool to produce environmental noise maps. *ISPRS International Journal of Geo-Information*, 8(3):130, 2019. doi: 10.3390/ijgi8030130.

Jérémy Gelb and Philippe Apparicio. Noise exposure of cyclists in Ho Chi Minh City: a spatio-temporal analysis using non-linear models. *Applied Acoustics*, 148:332–343, 2019. doi: 10.1016/j.apacoust.2018.12.031.

Chucri A. Kardous and Peter B. Shaw. Evaluation of smartphone sound measurement applications. *The Journal of the Acoustical Society of America*, 135(4):EL186–EL192, 2014. doi: 10.1121/1.4865269.

Stylianos Kephalopoulos, Marco Paviotti, Fabienne Anfosso-Lédée, Dirk Van Maercke, Simon Shilton, and Nigel Jones. Advances in the development of common noise assessment methods in Europe: the CNOSSOS-EU framework for strategic environmental noise mapping. *Science of The Total Environment*, 482–483:400–410, 2014. doi: 10.1016/j.scitotenv.2014.02.031.

Hanna Meyer and Edzer Pebesma. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12(9):1620–1633, 2021. doi: 10.1111/2041-210X.13650.

Ministry of Natural Resources and Environment of Viet Nam. QCVN 26:2010/BTNMT — national technical regulation on noise. Vietnamese national technical regulation, 2010.

Enda Murphy and Eoin A. King. Testing the accuracy of smartphones and sound level meter applications for measuring environmental noise. *Applied Acoustics*, 106:16–22, 2016. doi: 10.1016/j.apacoust.2015.12.012.

Quang Chinh Nguyen, An Thanh Thi Chu, Bao Gia Truong, and Duyen Hong Thuc Nguyen. Noise pollution in developing countries: loopholes and recommendations for Vietnam law. *City and Environment Interactions*, 25:100187, 2025a. doi: 10.1016/j.cacint.2025.100187.

Thu Lan Nguyen, Yuya Nishimura, and Sohei Nishimura. Horn use patterns and acoustic characteristics in congested urban traffic: a case study of Ho Chi Minh City. *Acoustics*, 7(2):36, 2025b. doi: 10.3390/acoustics7020036.

Solomon Nsumba et al. Urban Noise Uganda 61K: a smartphone-based urban noise dataset. *Scientific Data*, 2026. doi: 10.1038/s41597-026-06658-w.

Hai Yen Thi Phan, Takashi Yano, Tetsumi Sato, and Tsuyoshi Nishimura. Characteristics of road traffic noise in Hanoi and Ho Chi Minh City, Vietnam. *Applied Acoustics*, 71(5):479–485, 2010. doi: 10.1016/j.apacoust.2009.11.008.

David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendon A. Wintle, Florian Hartig, and Carsten F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. doi: 10.1111/ecog.02881.